

LAB EXPERIMENT RECORD

Set 4: Product Inventory Management

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze product inventory data using R programming by creating a bar chart, stacked bar chart, and scatter plot, and to develop a Tableau dashboard for interactive exploration of product quantities and category-wise inventory distribution.

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

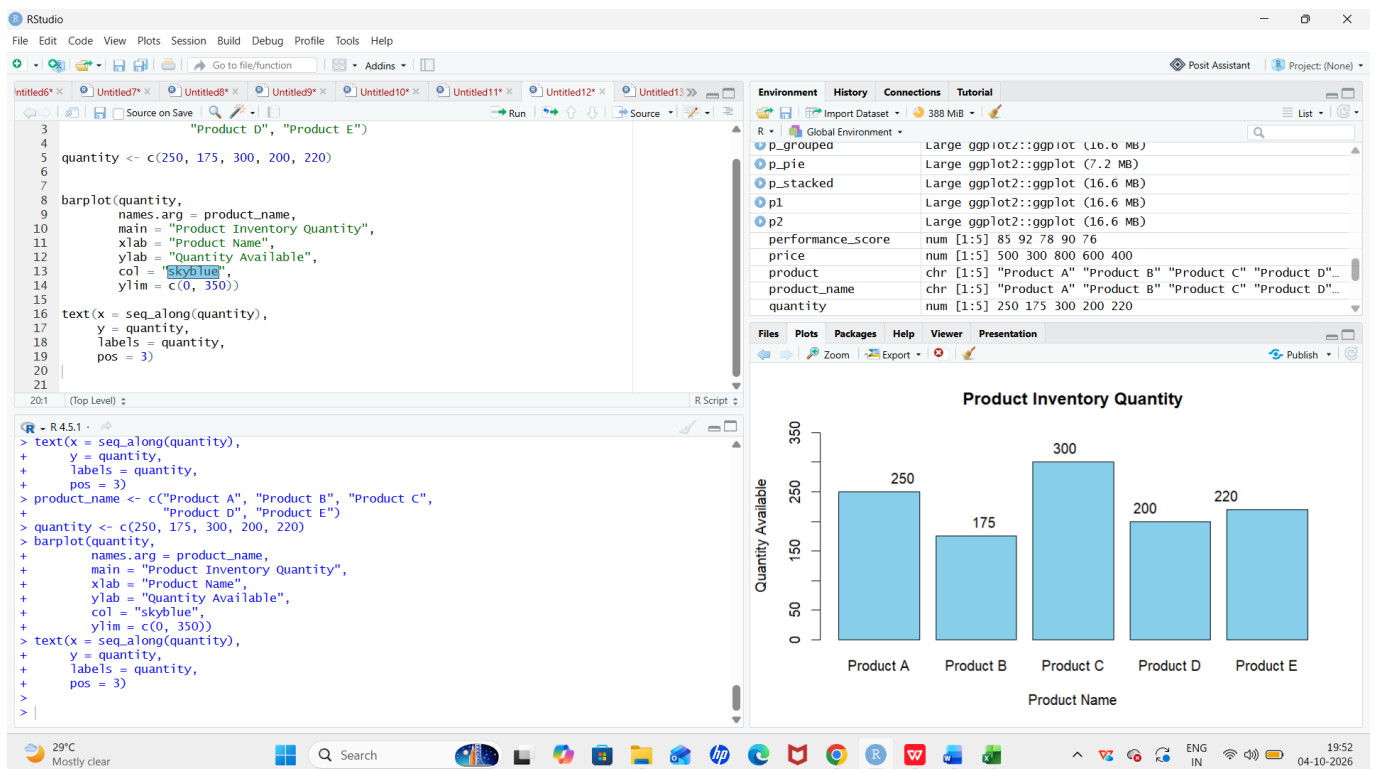
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title

```
product_name <- c("Product A", "Product B", "Product C",  
                  "Product D", "Product E")
```

```
quantity <- c(250, 175, 300, 200, 220)
```

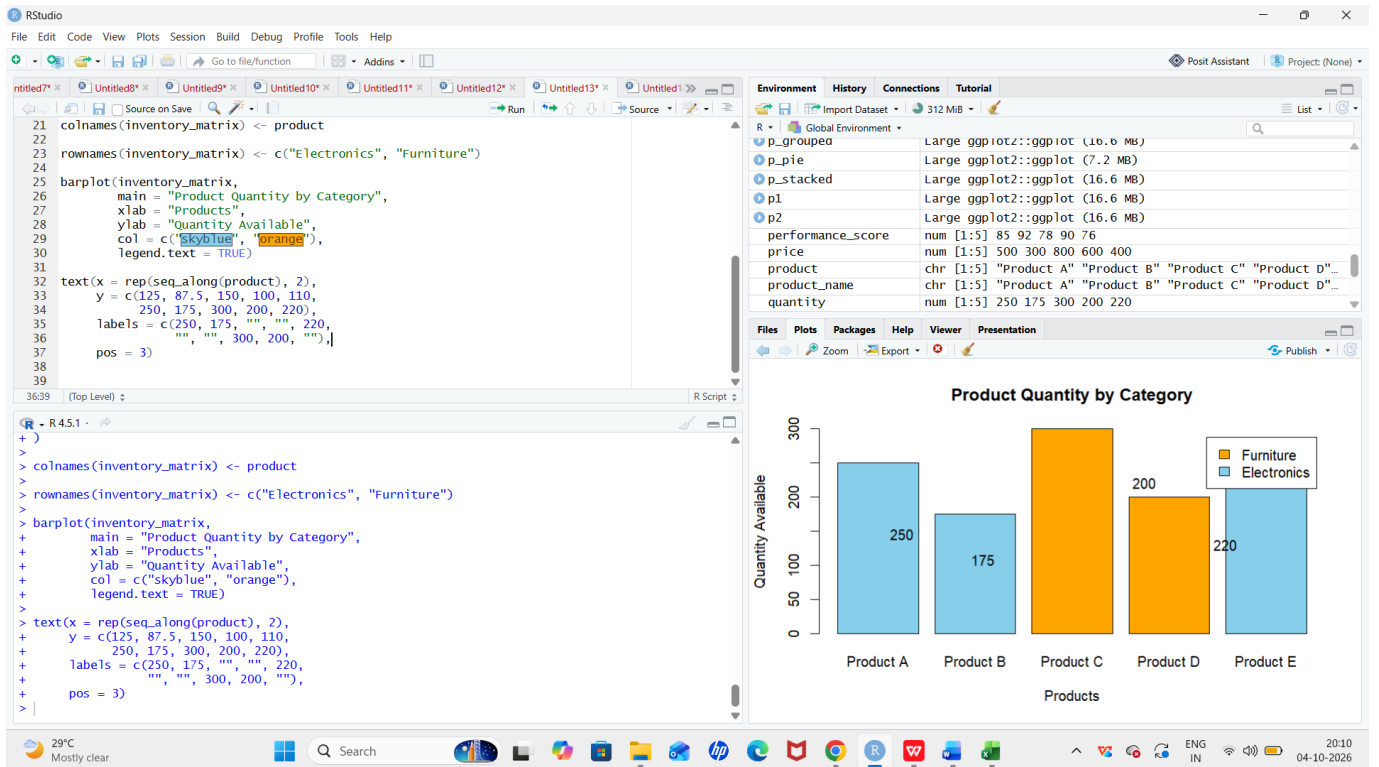
```
barplot(quantity,  
        names.arg = product_name,  
        main = "Product Inventory Quantity",  
        xlab = "Product Name",  
        ylab = "Quantity Available",  
        col = "skyblue",  
        ylim = c(0, 350))
```

```
text(x = seq_along(quantity),  
     y = quantity,  
     labels = quantity,  
     pos = 3)
```



2. Generate a stacked bar chart to show the quantity of each product within different product categories..

```
product <- c("Product A", "Product B", "Product C",  
            "Product D", "Product E")  
  
category <- c("Electronics", "Electronics",  
             "Furniture", "Furniture", "Electronics")  
  
quantity <- c(250, 175, 300, 200, 220)  
  
inventory <- data.frame(product, category, quantity)  
  
inventory_matrix <- matrix(  
  c(250, 175, 0, 0, 220,  
    0, 0, 300, 200, 0),  
  nrow = 2,  
  byrow = TRUE  
)  
  
colnames(inventory_matrix) <- product  
  
rownames(inventory_matrix) <- c("Electronics", "Furniture")  
  
barplot(inventory_matrix,  
        main = "Product Quantity by Category",  
        xlab = "Products",  
        ylab = "Quantity Available",  
        col = c("skyblue", "orange"),  
        legend.text = TRUE)  
  
text(x = rep(seq_along(product), 2),  
     y = c(125, 87.5, 150, 100, 110,  
           250, 175, 300, 200, 220),  
     labels = c(250, 175, "", "", 220,  
                "", "", 300, 200, ""),  
     pos = 3)
```



3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
product <- c("Product A", "Product B", "Product C",  
            "Product D", "Product E")
```

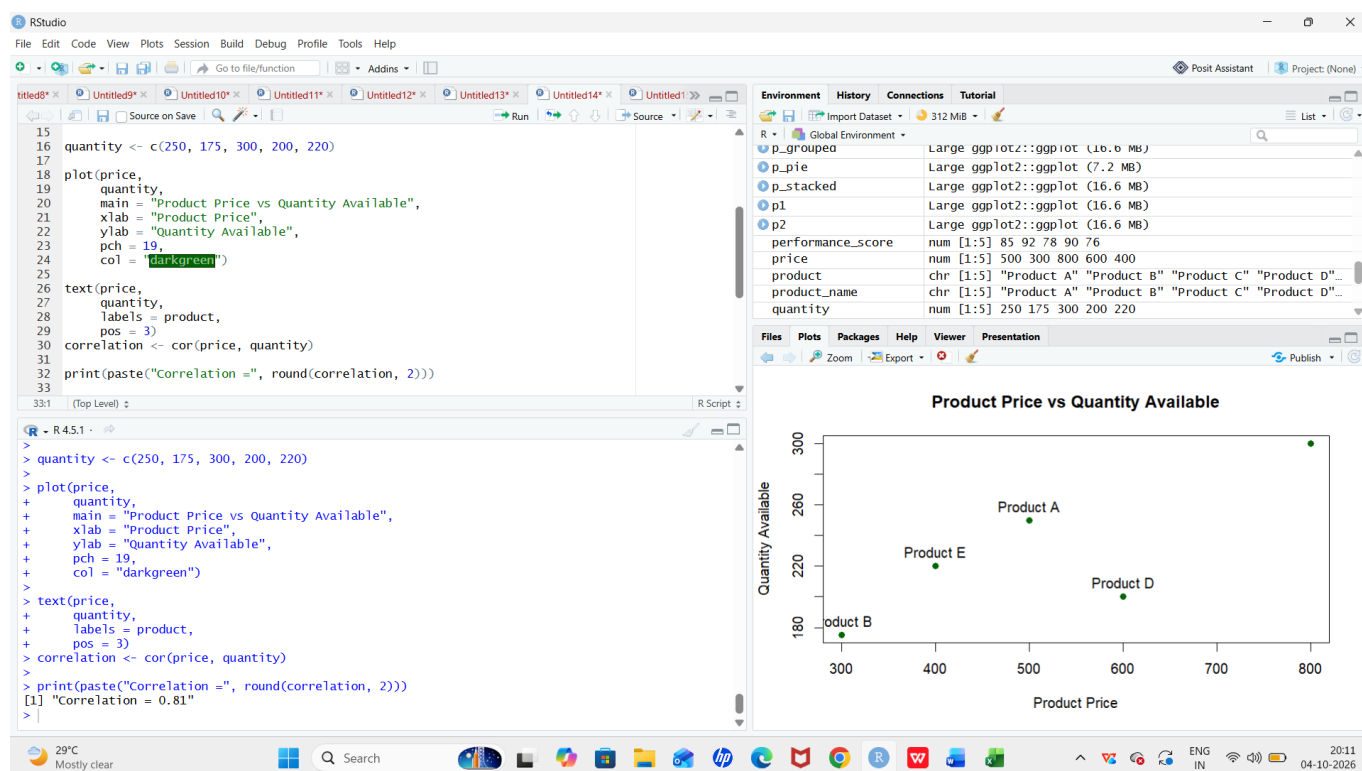
```
price <- c(500, 300, 800, 600, 400)
```

```
quantity <- c(250, 175, 300, 200, 220)
```

```
plot(price,  
      quantity,  
      main = "Product Price vs Quantity Available",  
      xlab = "Product Price",  
      ylab = "Quantity Available",  
      pch = 19,  
      col = "darkgreen")
```

```
text(price,  
      quantity,  
      labels = product,  
      pos = 3)  
correlation <- cor(price, quantity)
```

```
print(paste("Correlation =", round(correlation, 2)))
```



4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

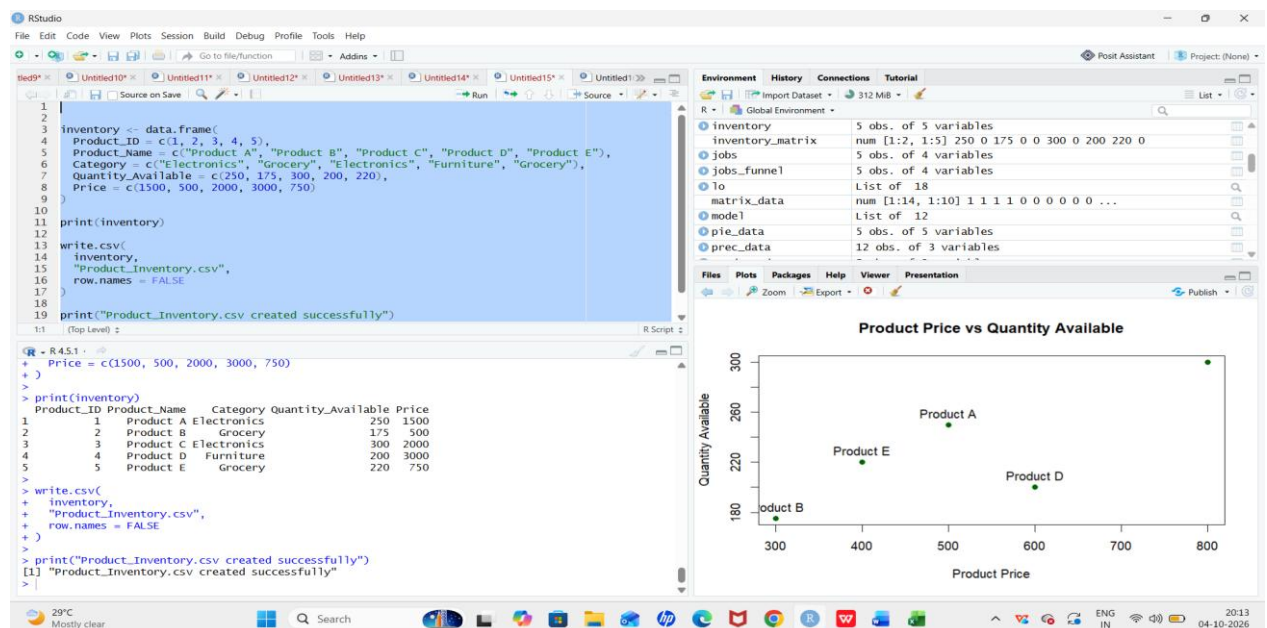
```
library(shiny)

inventory <- data.frame(
  Product_ID = c(1, 2, 3, 4,
5),
  Product_Name =
c("Product A", "Product
B", "Product C", "Product
D", "Product E"),
  Category =
c("Electronics", "Grocery",
"Electronics", "Furniture",
"Grocery"),
  Quantity_Available =
c(250, 175, 300, 200,
220),
  Price = c(1500, 500,
2000, 3000, 750)
)

print(inventory)

write.csv(
  inventory,
  "Product_Inventory.csv",
  row.names = FALSE
)

print("Product_Inventory.
csv created successfully"
)
```



Algorithm

1. Create the product inventory dataset in R with **Product ID, Product Name, Category, Quantity Available, and Price**.
2. Use the `barplot()` function to create a **bar chart** showing the quantity available for each product.
3. Add appropriate **axis labels and title** to the bar chart.
4. Organize the inventory data according to different product categories.
5. Use the `barplot()` function to create a **stacked bar chart** showing product quantities across different categories.
6. Create a **scatter plot** using product price and quantity available to analyze their relationship.
7. Add appropriate **axis labels, product names, and title** to the scatter plot.
8. Calculate the correlation between product price and quantity available using the `cor()` function.
9. Export the product inventory dataset as a **CSV file** using the `write.csv()` function.
10. Import the product inventory dataset into **Tableau**.
11. Create a **bar chart** in Tableau to display the quantity available for each product.
12. Create a **stacked bar chart** in Tableau to display the quantity of products according to their categories.
13. Combine both charts into a **Tableau dashboard** and add filters for interactive exploration.

Result

The product inventory data was successfully analyzed using **R programming**. The bar chart displayed the quantity available for each product, while the stacked bar chart showed the distribution of product quantities across different categories. The scatter plot helped analyze the relationship between product price and quantity available. Finally, the product inventory dataset was prepared and exported for use in **Tableau**, where a dashboard can be developed by combining the bar chart and stacked bar chart for interactive exploration.

